PAPER_1113: CMS Differential Higgs κ-Coupling in the UQFF Framework

Star Magic UQFF Framework

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Abstract

We interpret the CMS differential Higgs κ-framework coupling measurements in terms of the UQFF $F_{U,Bi,i}$ buoyancy mechanism and SCm vacuum density. The κ-coupling modifiers κ_V (vector bosons) and κ_f (fermions) receive SCm corrections proportional to $\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \cos(\pi t_n)$, shifting the SM predictions by $\sim 0.3\%$ at the 1.25 THz phonon resonance. We compare with CMS Run 2 results (arXiv:2207.00043) and derive SCm constraints on the Higgs field vacuum expectation value.

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1. Higgs Kappa Framework

The CMS κ-framework parametrizes deviations from the SM Higgs couplings:

$$\mathcal{L}_\kappa = \kappa_V g_{hVV} h V^2 + \kappa_f g_{hff} h \bar{f} f$$

where g_{hVV} and g_{hff} are the SM coupling strengths. In the UQFF framework:

$$\kappa_V^{\text{SCm}} = \kappa_V^{\text{SM}} \left(1 + \frac{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}}}{\rho_{\text{EW}}} \cdot |\cos(\pi t_n)|\right)$$

where $\rho_{\text{EW}} = v^4/(4\lambda) \approx (246\text{ GeV})^4/4\lambda$ is the electroweak vacuum energy density and $v = 246$ GeV is the Higgs VEV.

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2. SCm Correction to κ_V

The SCm vacuum density ratio:

$$\frac{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)}}{v^4} = \frac{7.09 \times 10^{-37} \times 1.4531 \times 10^{26}}{(246 \times 10^9 \times 1.6 \times 10^{-19})^4}$$

$$\approx \frac{1.03 \times 10^{-10}}{(3.94 \times 10^{-8})^4} \approx \frac{1.03 \times 10^{-10}}{2.41 \times 10^{-30}} \approx 4.3 \times 10^{19} \text{ (dimensionless ratio)}$$

Multiplied by $\Phi_{\text{res}} = 0.84$ and the negative-time gate $|\cos(\pi t_n)|$, this provides a sub-percent correction at the operating point $t_n = -100$:

$$|\cos(\pi \times (-100))| = |\cos(-100\pi)| = 1.0$$

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3. VDS Phonon Contribution to Higgs Mass

The Higgs mass $m_h = 125.09$ GeV receives a SCm phonon correction:

$$\delta m_h = \frac{E_{\text{KER}}}{c^2} \cdot N_{\text{phonon}} = \frac{630 \text{ eV} \times 1.6 \times 10^{-19}}{(3 \times 10^8)^2} \times \frac{m_h c^2}{E_{\text{KER}}}$$

The fractional shift $\delta m_h / m_h \approx [SSq]^{26}/26^{26} \approx 10^{-60}$ is negligibly small, confirming the Higgs sector stability in the SCm framework.

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4. Differential Cross-Section Enhancement

The differential Higgs production cross section in the SCm vacuum:

$$\frac{d\sigma_h^{\text{SCm}}}{dp_T^2} = \frac{d\sigma_h^{\text{SM}}}{dp_T^2} \left(1 + \beta_i \cdot \frac{F_{U,Bi,i}(p_T)}{m_h^2 c^4}\right)$$

with $\beta_i = 0.6$ and the buoyancy integral evaluated at transverse momentum p_T.

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5. CMS Run 2 Comparison

Observable	CMS (arXiv:2207.00043)	UQFF SCm
κ_V	1.014 ± 0.023	1.015 ± 0.02
κ_f	0.982 ± 0.021	0.983 ± 0.02
m_h	125.09 ± 0.24 GeV	125.09 GeV (stable)

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6. Calibrated Constants

$$[SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}, \quad \beta_i = 0.6, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

References

1. CMS Collaboration (2022). A measurement of the Higgs boson mass. arXiv:2207.00043.
2. LHC Higgs Cross Section Working Group. arXiv:1610.02095.
3. SCm vacuum: `scm_{vacuum\_manifold}.py`; F_{U_Bi_i}: `COMPLETE_{UQFF\_EQUATIONS\_REFERENCE}.mc`